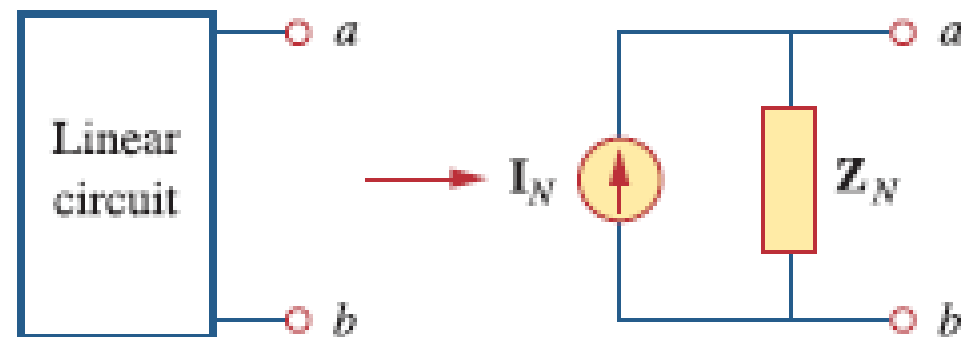


Norton's Theorem

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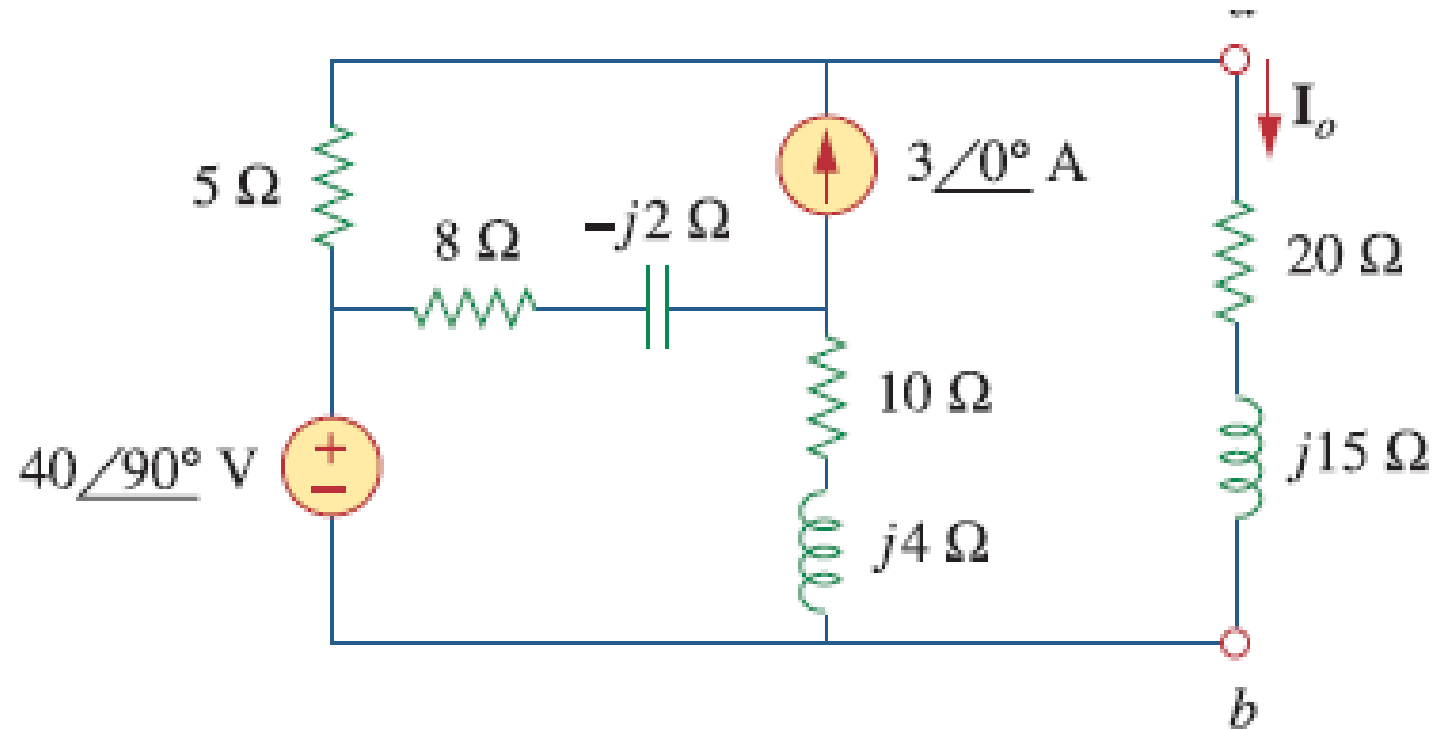
Norton's Theorem

Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N , where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.



Numerical -1

Obtain current $\mathbf{I}_o$ in Fig. using Norton's theorem.



Solution:

To find Z_N :

$$Z_N = 5 \Omega$$

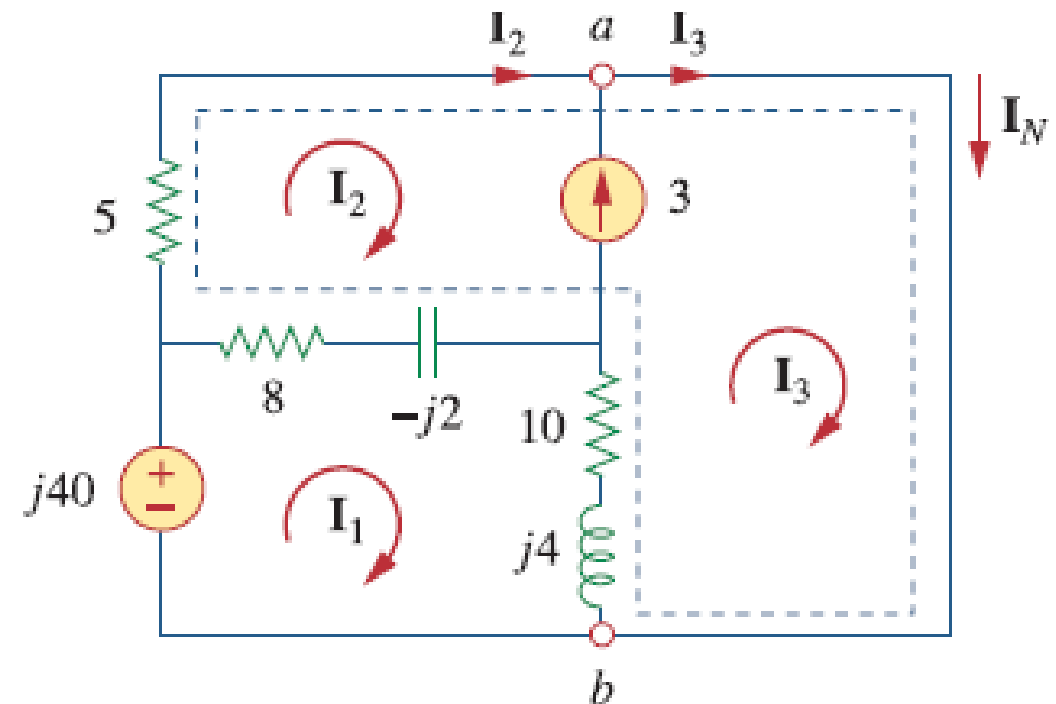
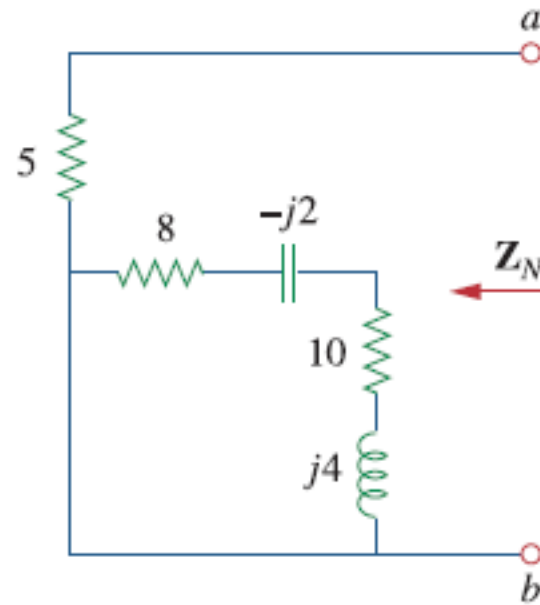
To find I_N :

For mesh 1,

$$-j40 + (18 + j2)\mathbf{I}_1 - (8 - j2)\mathbf{I}_2 - (10 + j4)\mathbf{I}_3 = 0$$

For the supermesh,

$$(13 - j2)\mathbf{I}_2 + (10 + j4)\mathbf{I}_3 - (18 + j2)\mathbf{I}_1 = 0$$



At node a ,

$$\mathbf{I}_3 = \mathbf{I}_2 + 3$$

Adding Mesh 1 and super mesh equations,

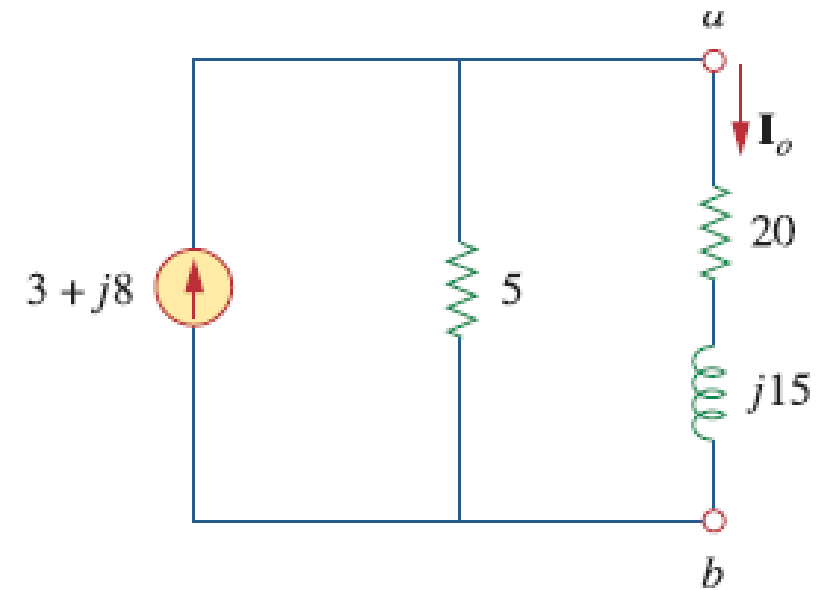
$$-j40 + 5\mathbf{I}_2 = 0 \quad \Rightarrow \quad \mathbf{I}_2 = j8$$

$$\mathbf{I}_3 = \mathbf{I}_2 + 3 = 3 + j8$$

The Norton current is

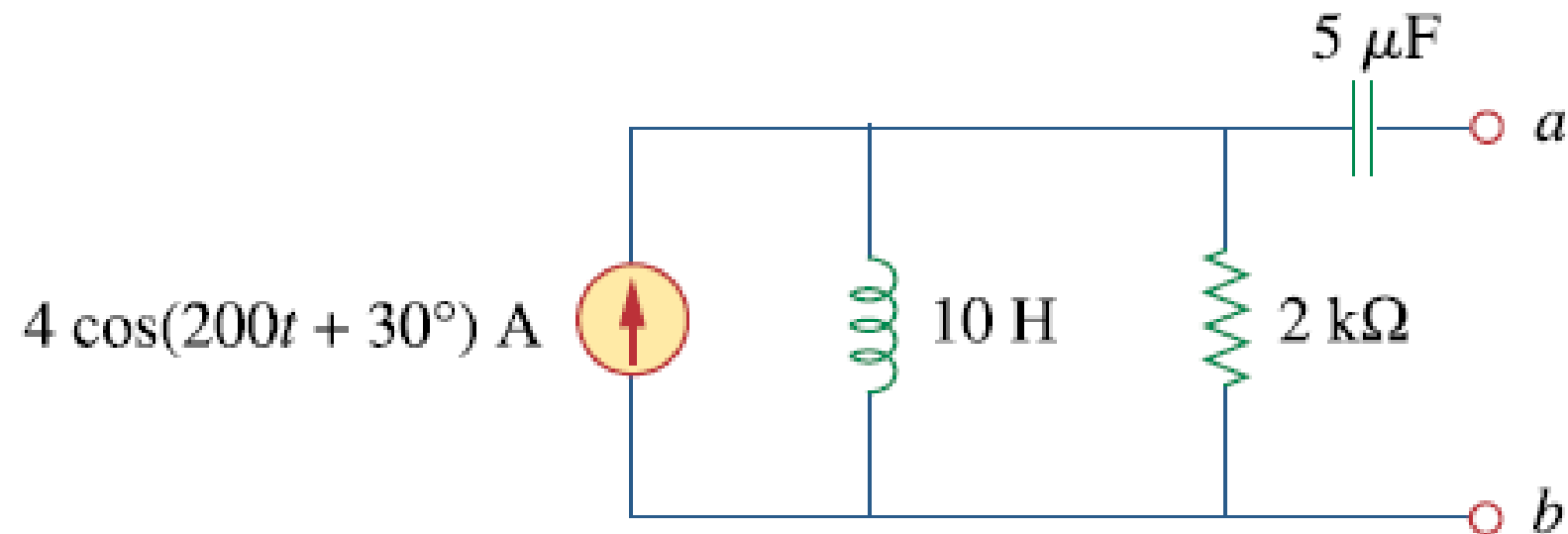
$$\mathbf{I}_N = \mathbf{I}_3 = (3 + j8) \text{ A}$$

$$\mathbf{I}_o = \frac{5}{5 + 20 + j15} \mathbf{I}_N = \frac{3 + j8}{5 + j3} = 1.465 \angle 38.48^\circ \text{ A}$$



Numerical -2

- Obtain the Norton's equivalent for the circuit.



- Solution:

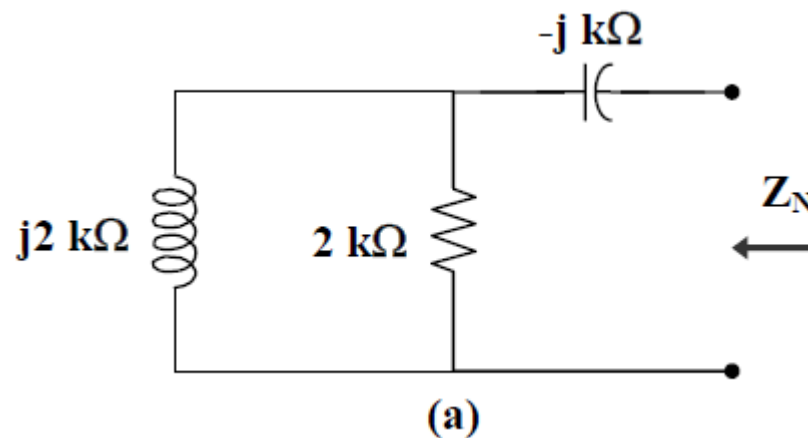
Transform the circuit to the frequency domain.

$$4 \cos(200t + 30^\circ) \longrightarrow 4 \angle 30^\circ, \quad \omega = 200$$

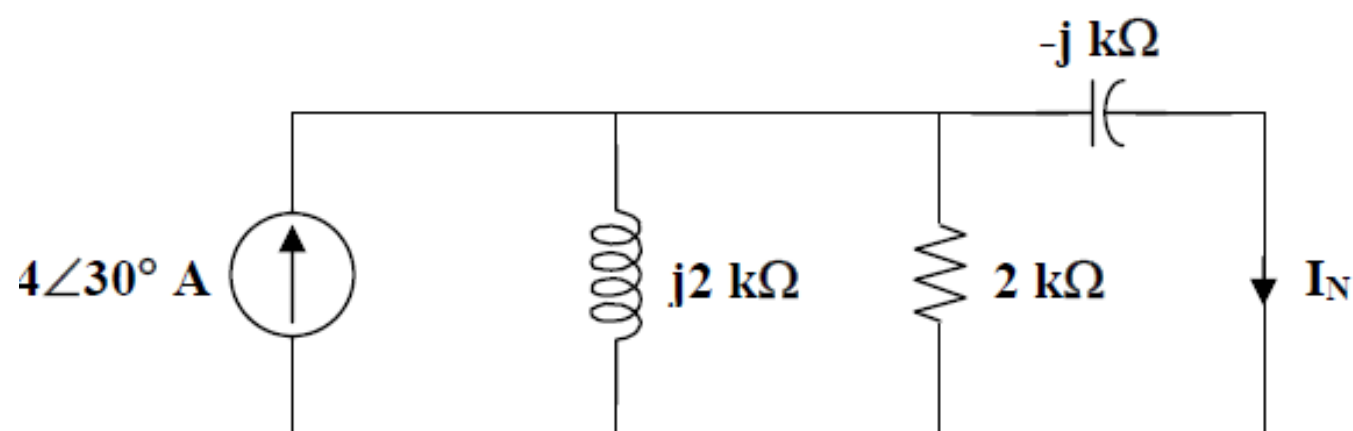
$$10 \text{ H} \longrightarrow j\omega L = j(200)(10) = j2 \text{ k}\Omega$$

$$5 \text{ }\mu\text{F} \longrightarrow \frac{1}{j\omega C} = \frac{1}{j(200)(5 \times 10^{-6})} = -j \text{ k}\Omega$$

$\mathbf{Z_N}$ is found using the circuit in Fig. (a).



$$\mathbf{Z_N} = -j + 2 \parallel j2 = -j + 1 + j = 1 \text{ k}\Omega$$



$$j2 \parallel 2 = 1 + j$$

By the current division principle,

$$\mathbf{I}_N = \frac{1 + j}{1 + j - j} (4\angle 30^\circ) = 5.657\angle 75^\circ$$

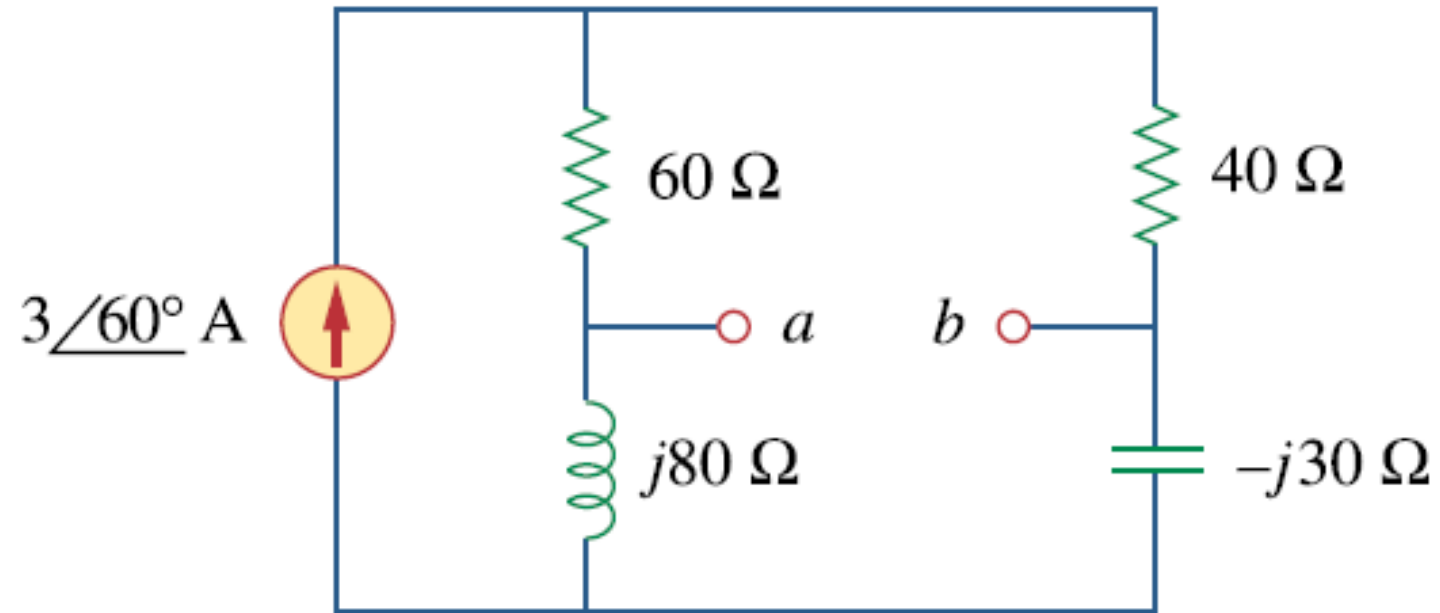
Therefore,

$$i_N = \underline{\underline{5.657 \cos(200t + 75^\circ) \text{ A}}}$$

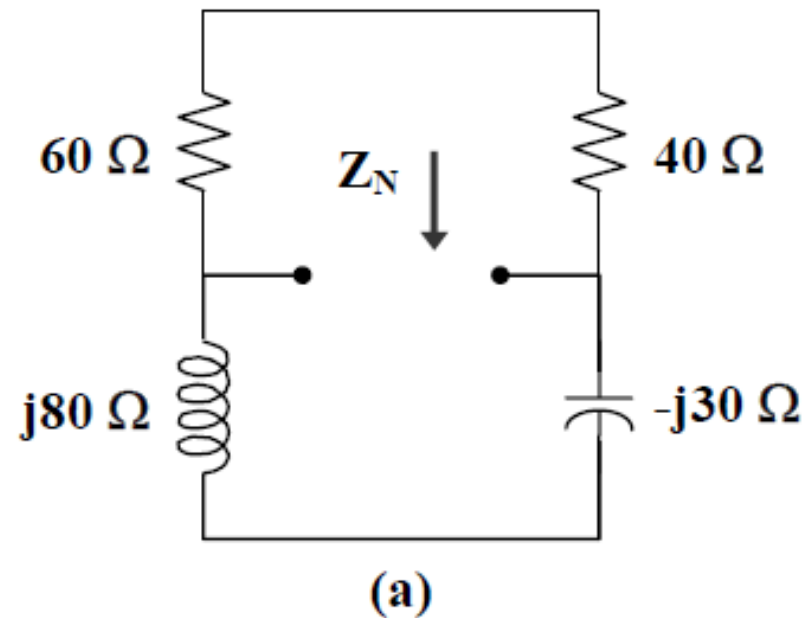
$$\mathbf{Z}_N = \underline{\underline{1 \text{ k}\Omega}}$$

Numerical -3

- Obtain the Norton's equivalent for the circuit.



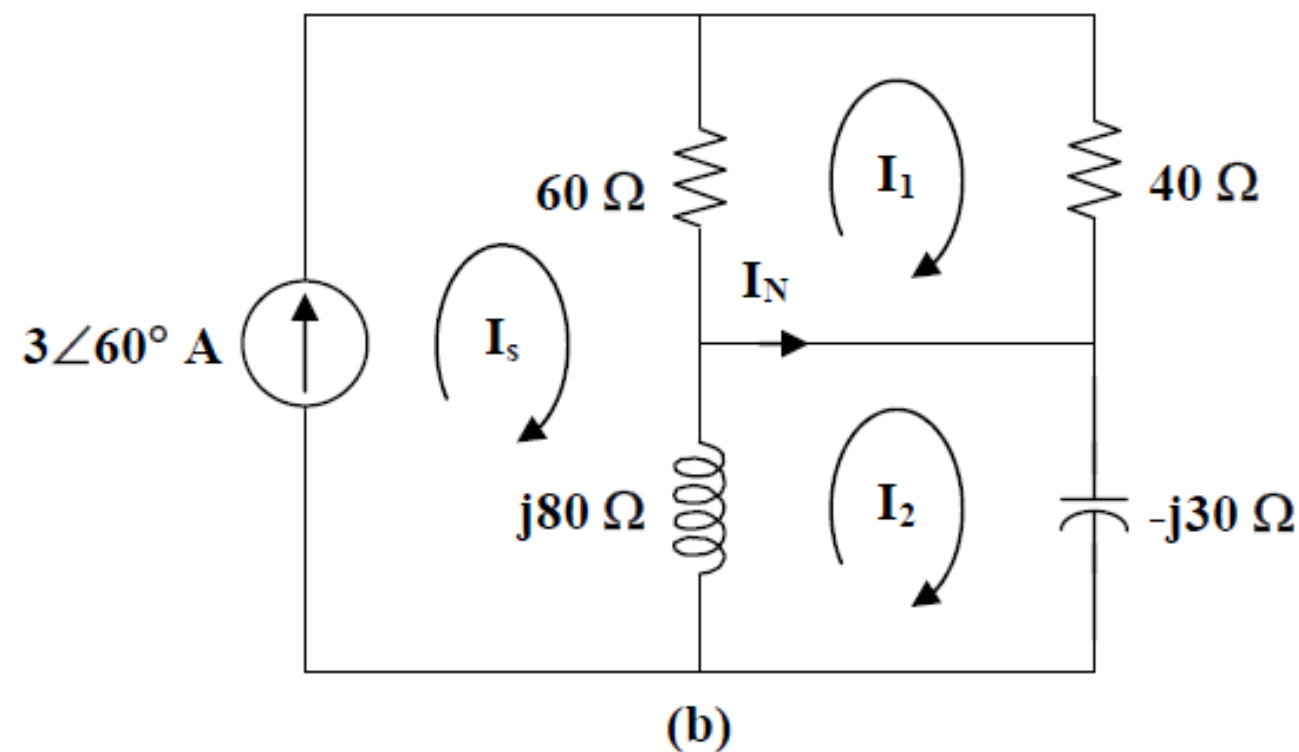
- Solution: Z_N is obtained from the circuit in Fig. (a).



$$Z_N = (60 + 40) \parallel (j80 - j30) = 100 \parallel j50 = \frac{(100)(j50)}{100 + j50}$$

$$Z_N = 20 + j40 = \underline{44.72 \angle 63.43^\circ \Omega}$$

To find $\mathbf{I}_N$, consider the circuit in Fig. (b).



$$\mathbf{I}_s = 3\angle 60^\circ$$

For mesh 1,

$$100\mathbf{I}_1 - 60\mathbf{I}_s = 0$$

$$\mathbf{I}_1 = 1.8\angle 60^\circ$$

For mesh 2,

$$(j80 - j30)\mathbf{I}_2 - j80\mathbf{I}_s = 0$$

$$\mathbf{I}_2 = 4.8\angle 60^\circ$$

$$\mathbf{I}_N = \mathbf{I}_2 - \mathbf{I}_1 = \underline{\underline{3\angle 60^\circ \text{ A}}}$$